

Program: Diploma in Food Processing Technology	
Course Code: 3427	Course Title: Elements of Food Engineering Lab
Semester: 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- The student will acquaint knowledge for performing various techniques in determining properties such as density, volume, porosity, coefficient of friction, angle of repose etc
- Develop skills to operate equipments for size reduction, separation etc.

Course Prerequisites:

Topic	CourseCode	Course Name	Semester
Physical aero and hydrodynamic properties		Applied physics	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Practice different techniques for determining the physical properties of food such as density, volume, porosity, surface area, shape and size	18	Applying
CO2	Practice different techniques for determining the frictional properties of food such as coefficient of friction, angle of repose etc	8	Applying
CO3	Explain and use equipments for size reduction, conveying and separation of food materials	3	Applying
CO4	Design and carry out scientific experiments as well as accurately record and analyze the results of such experiments	17	Applying

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3	3		
CO2				3	3		
CO3				3	3		
CO4				3	3		

3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Practice different techniques for determining the physical properties of food such as density, volume, porosity, surface area, shape and size		
M1.01	Determination of firmness of fruits and vegetables	3	Understanding
M1.02	Determination of size and shape of various food products	3	Applying
M1.03	Determination of volume, true density and bulk density of various food products.	3	Applying
M1.04	Study on particle size distribution of various food grains using sieve analysis.	3	Applying
M1.05	Estimation of surface area of fruits and vegetables.	3	Applying
M1.06	Determination of porosity of foods using air pycnometer	3	Applying
CO2	Practice different techniques for determining the frictional properties of food such as coefficient of friction, angle of repose etc		
M2.01	Determination of efficiency of separation of various food products using inclined belt separator.	3	Applying

M2.02	Determination of mixing index of different grains	3	Applying
	Lab Test –I	2	
CO3	Explain and use equipments for size reduction, conveying and separation of food materials		
M3.01	Experiment on size reduction of solid foods using ball mill	3	Applying
CO4	Design and carry out scientific experiments as well as accurately record and analyse the results of such experiments		
M4.01	Determination of thermal conductivity of foods.	3	Applying
M4.02	Determination of specific heat of solid foods	3	Applying
M4.03	Determination of viscosity of foods by rotational viscometer	3	Applying
M4.04	Study on bucket elevator and determination of optimum capacity	3	Applying
M4.05	Study on experiment on ohmic heating	3	Applying
	Lab Test– II	1	

Text/Reference

T/R	BookTitle/Author
R1	Rao M.A. and Rizvi, S.S.H. "Engineering Properties of Foods", Marcel Dekker, New York, 1986.
R2	Sergio O Serna Saldivar, Cereal grains laboratory references and procedure manual, Taylor and Francis group, 2012
R3	Nuri N Mohsenin, Physical properties of plant and animal materials, second edition, Gordon and research science publishers, 1986

Online Resources

Sl. No	Website Link
1.	https://thermtest.com/what-is-thermal-conductivity
2.	http://hyperphysics.phy-astr.gsu.edu/hbase/thermo/spht.html
3.	https://www.merriam-webster.com/dictionary/porosity
4	https://www.britannica.com/science/density